Let's refresh our memory about the customary units of measurement like inches, feet, yards, cups, pounds and more.

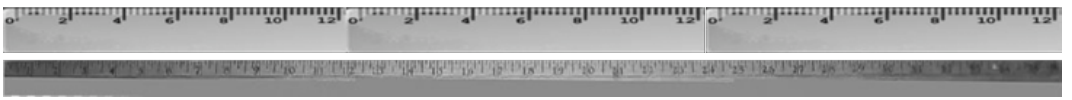
This is a customary ruler for measuring length:

12 inches equals a foot or

12 in = 1 ft

3 ft equals 1 yard or

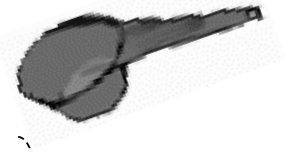
3 ft = 1 yd



36 inches = 1 yard (yd)

Units for measuring capacity:

1 ounce (oz) is less than (<) 1 cup(c)



1 c is < 1 pint (pt)

1 quart(qt)<1gallon(gal)

Units for measuring mass:

1 ounce (1 oz) <

1 pound (1 lb) <

1 ton

Let's practice what we know:

16 ozs = 1 lb

So if you are going to measure a dozen thumbtacks, you would use the unit

ounces instead of pounds.

To measure the quantity of meat to buy for a family of four, you would use

pounds, not tons.

From smallest to greatest the capacity measurements range from:

Fluid Ounces Cups Pints

Quarts Gallons

1 oz	◯	1 lb
1 ton	◯	1 oz
1 c	◯	1 qt
1 gal	◯	1 qt
1 qt	◯	1 pt
1 ft	◯	1 yd
1 in	◯	1 yd
1 yd	◯	1 ft
1 gal	◯	1 c

Comparing Units

Trace the dotted words and numbers. Use the symbols < or > to compare each unit.

1 oz ◯ < 1 lb

1 ton ◯ > 1 oz

1 c ◯ < 1 qt

1 gal ◯ > 1 qt

1 qt ◯ > 1 pt

1 ft ◯ < 1 yd

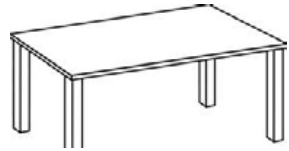
1 in ◯ < 1 yd

1 yd ◯ > 1 ft

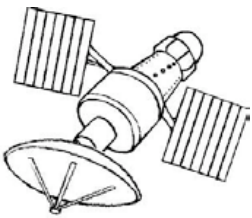
1 gal ◯ > 1 c

Comparing Units *Answers*

To measure the length of a table, you would need to use a ruler. But the ruler would need to be longer than 12 inches, so a yardstick with 36 inches would work.



If you were measuring the length of a long play on a football field, you would use yards.



If you were measuring the distance to the moon or a satellite you would use miles, not yards.

To measure the weight or mass of a truck or ship, you would use tons, not pounds.

Ice cream can be bought in gallon containers, but if you were to fix you an ice cream cone, you would use a cup to measure it.

If you were going to measure the weight of a handful of paper clips, you would use ounces.

Which unit would describe a quantity of water you would add to a saucepan?

2 cups or 2 pounds

What unit would you use to measure the height of a drinking glass?

8 inches or 8 yards

Which is the better estimate for the height of a tall mountain?

3 yards or 3 miles

For a charity-walk, what would be the most logical distance to cover?

5 feet or 5 miles

Practicing Identifying Reasonable Units

Trace the dotted words and numbers. Draw a circle around the correct answer.

Which unit would describe a quantity of water you would add to a saucepan?

2 cups or 2 pounds

What unit would you use to measure the height of a drinking glass?

8 inches or 8 yards

Which is the better estimate for the height of a tall mountain?

3 yards or 3 miles

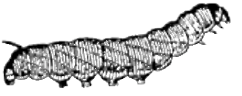
For a charity-walk, what would be the most logical distance to cover?

5 feet or 5 miles

Let's refresh our memory about the metric units of measurement like meters, liters and grams. A conversion chart such as this will help: (from smallest to largest)

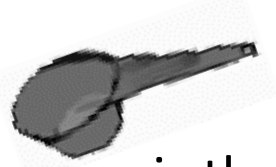
milli	centi	deci	unit	deca	hecto	kilo
Milligram	**Centigram**	**Decigram**	**gram**	**decagram**	**hectogram**	**Kilogram**

✱The metric unit for measuring length is the meter.

A millimeter is smaller  than a kilometer. A caterpillar would have a length measured in millimeters, but a distance from one city to another would best be measured in kilometers.

Metric Unit Chart

✱The metric unit for measuring capacity is the liter. A milliliter is smaller than a liter. Soda is often sold in a liter bottle, but a small dose of medicine would best be measured in milliliters.



✱The metric unit for measuring mass is the gram. A gram is smaller than a kilogram. The weight of a box of crackers would be listed on the box in grams, but a person's weight would best be measured in kilograms. A thumbtack would be measured in milligrams.

✓ Milligram (mm) gram (g)
kilogram (kg)

✓ liter (l) milliliter (ml)
kiloliter (kl)

✓ meter (m) kilometer(km)
millimeter (mm)

1,000 mg = 1 g

1,000 g = 1 kg

1,000,000 mg = 1 kg

1,000 ml = 1 l

milligram	◯	gram
liter	◯	milliliter
kilometer	◯	meter
gram	◯	milligram
milliliter	◯	kiloliter
meter	◯	millimeter
milligram	◯	kilogram
liter	◯	kiloliter
kilometer	◯	millimeter
gram	◯	kilogram

Metric Units

numbers. Put > (greater than) or < (less than) in each circle.

milligram ◯ < gram

liter ◯ > milliliter

kilometer ◯ > meter

gram ◯ > milligram

milliliter ◯ < kiloliter

meter ◯ > millimeter

milligram ◯ < kilogram

liter ◯ < kiloliter

kilometer ◯ > millimeter

gram ◯ < kilogram

The units of length in the customary system are inches, feet, yards and miles.

The table at the right shows the relationship between these units.

Customary Units of Length
1 mile (mi) = 5,280 feet (ft)
1 foot (ft) = 12 inches (in)
1 yard (yd) = 3 feet (ft)
1 yard (yd) = 36 inches (in)

- To convert from larger units to smaller units, multiply.
- To convert from smaller units to larger units, divide.

9 ft = 9 X 12 = 108 in

7 yd = 7 X 36 = 252 in

There will be a greater number of smaller units than larger units.

60 in = 60 ÷ 12 = 5 ft

81 ft = 81 ÷ 3 = 27 yd

There will be fewer larger units than smaller units.

Converting Customary Units

Trace the dotted words.

The units of weight in the customary system are ounce, pound, and ton. The table at the right shows the relationship between these units. As with units of length, to convert from larger units to smaller units, you multiply.
To convert from smaller units to larger units, you divide.

Customary Units of Weight
1 pound (lb) = 16 ounces (oz)
1 ton (T) = 2,000 pounds (lb)

6 lbs = 6 X 16 = 96 oz

12, 400 lb = 12400 ÷ 16 = 6.2 T

9 T = 9 X 2000 = 18,000 lbs

Capacity is the amount of liquid or dry substance a container can hold.

Customary units of capacity are fluid ounce, cup, pint, quart, and gallon. The relationships among these units are shown in the table to the right.

Customary Units of Capacity
1 cup (c) = 8 fluid ounces (fl oz)
1 pint (pt) = 2 cups
1 quart (qt) = 2 pints
1 gallon (gal) = 4 quarts

6 gal = 6 X 4 = 24 qts

10 pts = 10 ÷ 2 = 5 qt

Customary Units of Length
1 mile (mi) = 5,280 feet (ft) 1 foot (ft) = 12 inches (in) 1 yard (yd) = 3 feet (ft) 1 yard (yd) = 36 inches (in)

13 ft. = 13 X 12 = ____ in

36 in. = 36 ÷ 12 = ____ ft

20 yd. = 20 X 36 = ____ in

Customary Units of Weight
1 pound (lb) = 16 ounces (oz) 1 ton (T) = 2,000 pound (lb)

18 lbs = 18 X 16 = ____ oz

32, 000 lb = 32000 ÷ 2000 = ____T

Multiply or divide to find the correct answer.

Customary Units of Length
1 mile (mi) = 5,280 feet (ft) 1 foot (ft) = 12 inches (in) 1 yard (yd) = 3 feet (ft) 1 yard (yd) = 36 inches (in)

13 ft. = 13 X 12 = _____ in 156 in

36 in. = 36 ÷ 12 = _____ ft 3 ft

20 yd. = 20 X 36 = _____ in 720 in

Customary Units of Weight
1 pound (lb) = 16 ounces (oz) 1 ton (T) = 2,000 pound (lb)

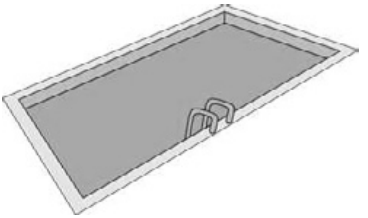
18 lbs = 18 X 16 = _____ oz 288 oz

32, 000 lb = 32000 ÷ 2000 = _____T 16 T

In review: The length of a bedroom would be

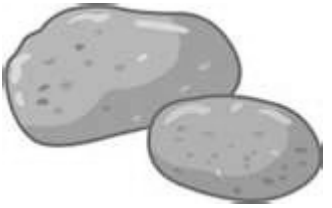
best measured in meters.

The volume of the water in a swimming pool would be best



measured in kiloliters.

The weight of a bag of potatoes would be best measured in



kilograms, but a small bag of

marbles in grams.

Use what you have just reviewed to add mixed measures.

Find the sum of 7 feet 4 inches and 2 feet 9 inches, then simplify.

7 ft 4 in Line up like units and add.

\+ 2 ft 9 in

9 ft **13 in.** = 9 ft + **(12 in. + 1 in.)**

Separate 13 in. into 12 in. and 1 in.

= 10 ft + 1 in. Replace 12 in. with 1 ft and add like units.

Add this one:

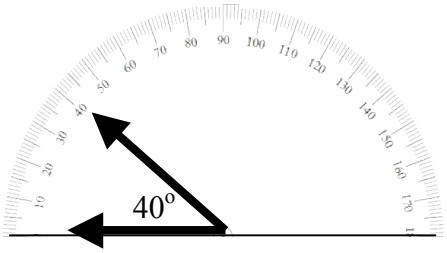
4 ft 1 in

\+ 3 ft 11 in

7 ft 12 in = 8 ft

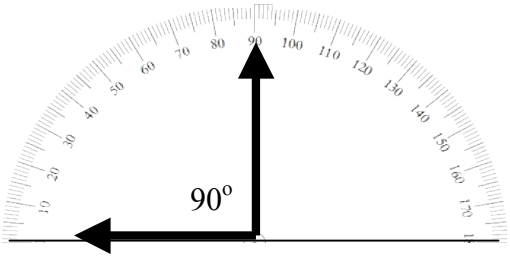
4 c 6 fl oz + 1 c 4 fl oz 5 c 10 fl oz	•	• 12 c 4 fl oz
7 c 5 fl oz + 4 c 7 fl oz 11 c 12 fl oz	•	• 6 c 2 fl oz
4 ft 11 in +12 ft 11 in 16 ft 22 in	•	• 11 ft 2 in
8 ft 7 in + 2 ft 7 in 10 ft 14 in	•	• 17 ft 10 in

4 c	6 fl oz	
+ 1 c	4 fl oz	
5 c	10 fl oz	12 c 4 fl oz
		6 c 2 fl oz
7 c	5 fl oz	
+ 4 c	7 fl oz	
11 c	12 fl oz	
4 ft	11 in	
+12 ft	11 in	11 ft 2 in
16 ft	22 in	
		17 ft 10 in
8 ft	7 in	
+ 2 ft	7 in	
10 ft	14 in	



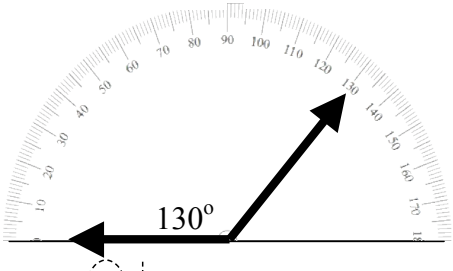
Acute Angle

Angle < 90°



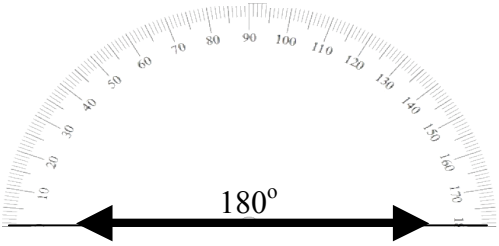
Right Angle

Angle = 90°

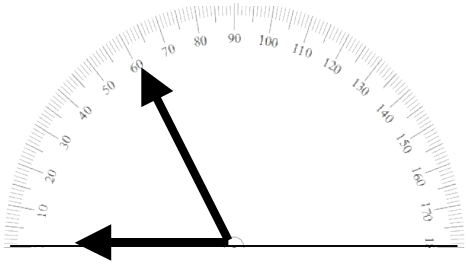


Obtuse Angle

Angle > 90°

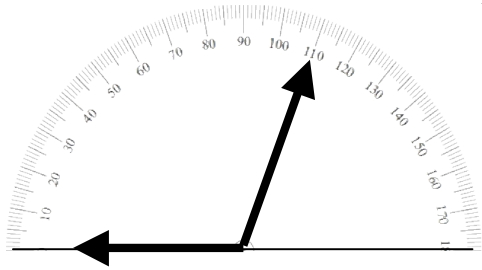


Line Angle = 180°



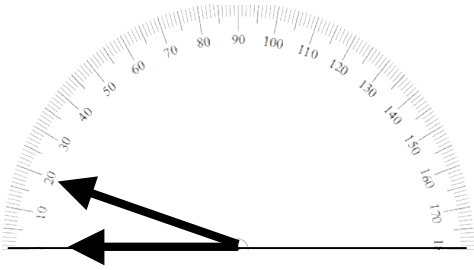
Acute Angle

Angle = 60°



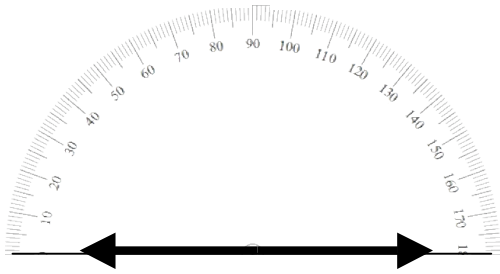
Obtuse Angle

Angle = 110°



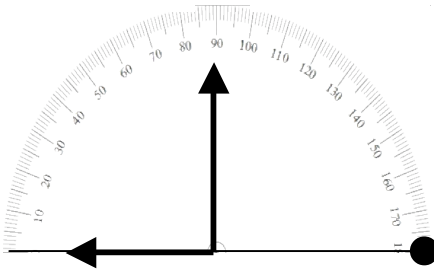
Acute Angle

Angle = 20°

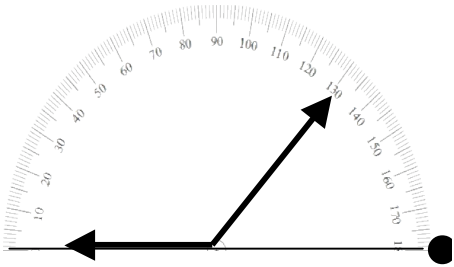


Line

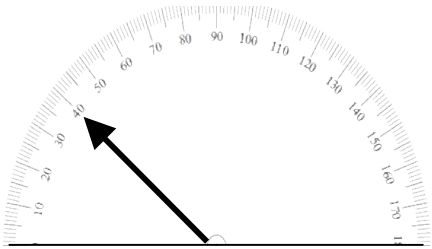
Angle = 180°



- Obtuse



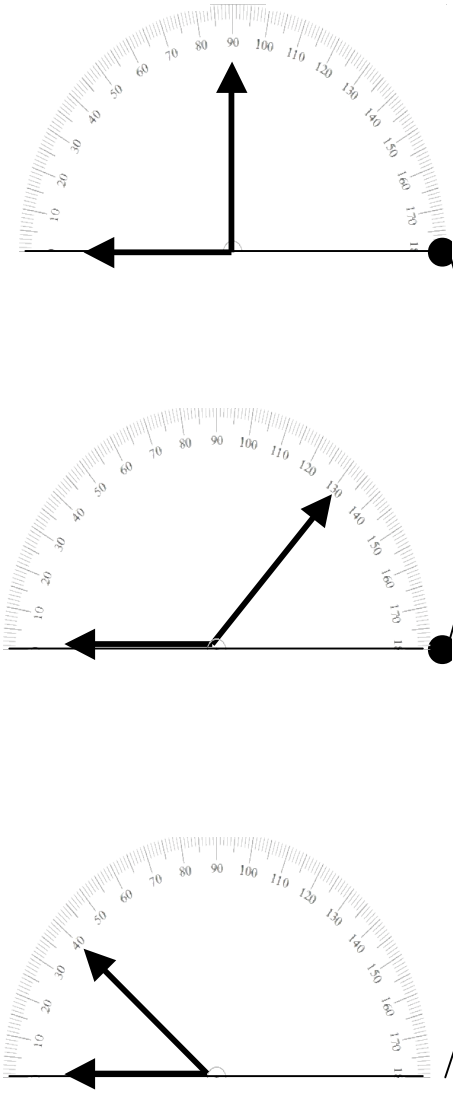
- Acute



- Right

Angle Identification Practice

from the correct picture of each angle to the correct name of the angle.

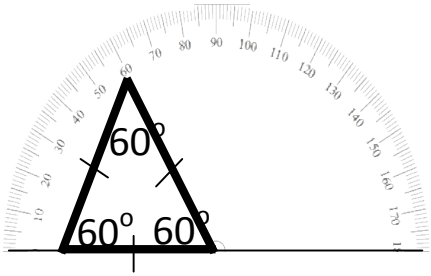


Right

Acute

Obtuse

A triangle has three sides and three angles. The three angles always add up to 180°.



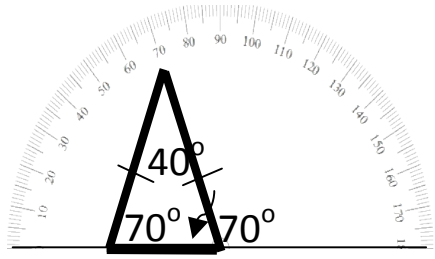
Equilateral Triangle

Three equal sides

Three equal angles,

always 60°

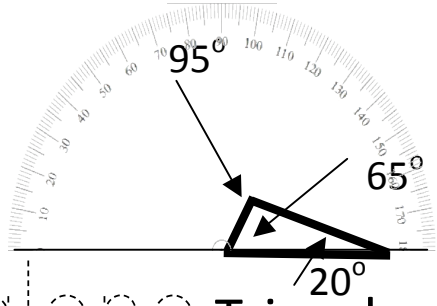
$60^o + 60^o + 60^o = 180^o$



Isosceles Triangle

Two equal sides

Two equal angles



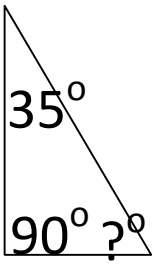
Scalene Triangle

No equal sides or angles

If you know the measure of two angles, you can use addition and subtraction to find the measure of the missing angle. Add the two angles known & subtract that sum from 180°.

$180^o - (90^o + 35^o) =$

$180^o - 125^o = 55^o$



Angle 1 = 25°

Angle 2 = 80°

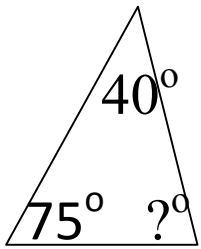
Angle 3 = $?^o$

$180^o - (25^o + 80^o)$

$180^o - 105^o = 75^o$

$180^o - (75^o + 40^o) =$

$180^o - 115^o = 65^o$



Angle 1 = 44°

Angle 2 = 96°

Angle 3 = $?^o$

$180^o - (44^o + 96^o)$

$180^o - 140^o = 40^o$

Angle 1 = 71°
Angle 2 = 32°
Angle 3 = ?°

180°-(71°+32°)

180° - 103° =____°

Angle 1 = 11°
Angle 2 = 119°
Angle 3 = ?°

180°-(11°+119°)

180° - 130° = ____°

Angle 1 = 64°
Angle 2 = 15°
Angle 3 = ?°

180°-(64°+15°)

180° - 79° =____°

Angle 1 = 44°
Angle 2 = 56°
Angle 3 = ?°

180°-(44°+56°)

180° - 100° = ____°

Angle 1 = 94°
Angle 2 = 21°
Angle 3 = ?°

180°-(94°+21°)

180° - 115° =____°

Angle 1 = 38°
Angle 2 = 22°
Angle 3 = ?°

180°-(38°+22°)

180°- 60° = ____°

Angle 1 = 71°
Angle 2 = 32°
Angle 3 = ?°

180°-(71°+32°)

180° - 103° = 77°

Angle 1 = 11°
Angle 2 = 119°
Angle 3 = ?°

180°-(11°+119°)

180° - 130° = 50°

Angle 1 = 64°
Angle 2 = 15°
Angle 3 = ?°

180°-(64°+15°)

180° - 79° = 101°

Angle 1 = 44°
Angle 2 = 56°
Angle 3 = ?°

180°-(44°+56°)

180° - 100° = 80°

Angle 1 = 94°
Angle 2 = 21°
Angle 3 = ?°

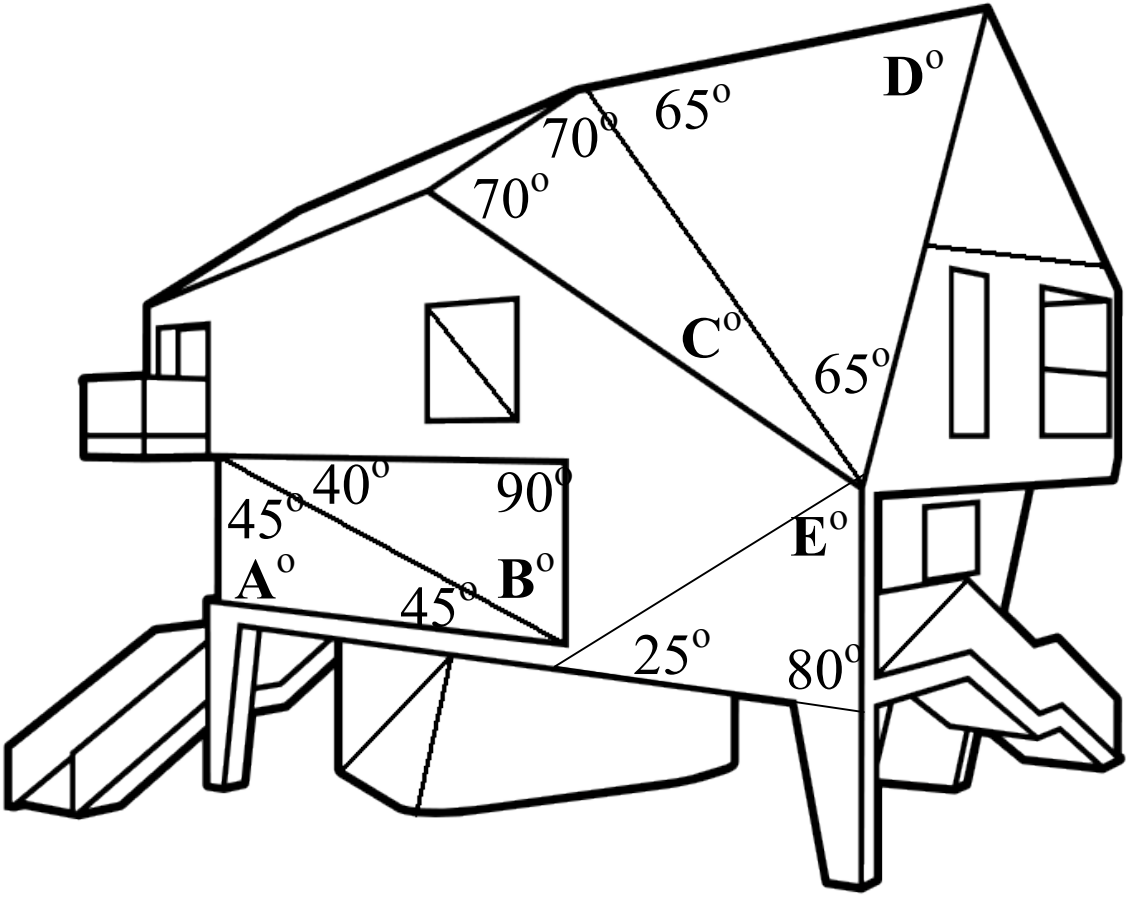
180°-(94°+21°)

180° - 115° = 65°

Angle 1 = 38°
Angle 2 = 22°
Angle 3 = ?°

180°-(38°+22°)

180° - 60° = 120°



$180^\circ - (65^\circ + 65^\circ) = 50^\circ$ ●	● A
$180^\circ - (70^\circ + 70^\circ) = 40^\circ$ ●	● B
$180^\circ - (45^\circ + 45^\circ) = 90^\circ$ ●	● C
$180^\circ - (25^\circ + 80^\circ) = 75^\circ$ ●	● D
$180^\circ - (90^\circ + 40^\circ) = 50^\circ$ ●	● E

Missing Measurement

missing measurement. Draw a line from the equation to the correct letter identifying it.